

Technical Appendix

Reproducibility Scope and Data

The eight primary public benchmarks are AgentRewardBench, AgentProcessBench, HINTBench, TraceElephant, CodeTraceBench, OSWorld-Human, AgentBoard, and the TraceView/ASE action corpus. Mind2Web and ScienceWorld provide two additional field-mapping checks. No novel benchmark dataset is introduced. Public benchmark targets, human stages, outcomes, and scores are used only by scorers after each method’s output is fixed.

The paper’s theoretical contribution is the semantic operation stack model, specified by the formal definitions in the Design section. AGENTPROF is a 9.8K-LOC implementation. Its executable source, preprocessing scripts, and released-data manifests accompany the paper in the code-and-data package.

Automatic Segmentation and Identity

The direct annotation backend is the OpenAI Codex CLI (version 0.145.0, model `gpt-5.6-sol`, high reasoning effort, default decoding, sandboxed non-interactive mode). Each worker receives one source-only trajectory packet, makes one annotation call with at most one format retry, and shares no state with other workers. Prompt and response previews are truncated to 1,800 characters, tool commands to 600, and status previews to 200; additive measures come from structured counters and are not truncated. The fixed instruction is reproduced in the main paper. It emits sparse path changes whose names begin with a verb and contain one to three words; intervening steps inherit the preceding path.

Operationally, each call is one isolated backend request. Here, recursion refers to the variable-depth nested decomposition constructed by the annotation agent within that request, rather than to a sequence of backend requests. “Iteratively” refers to revising the mark set before returning one final sparse annotation file.

Before folding, one fixed source-only action–object map canonicalizes free-form interval names. The map is implemented as ordered verb, object, and qualifier phrase tables with fixed aliases and stop words. It reads operation names and sparse source-only marks, independent of task IDs, benchmark names, outcomes, target stages, and score rows. Unmatched source words remain available to the boundary-safe refinement. If canonicalization would make adjacent display paths equal, the refinement retains the minimum source words needed to distinguish them; unresolved collisions fail closed. Equal canonical names receive one stable operation ID across sessions. On the full CodeTraceBench population, the map covers all 3,895 free-form names, applies 503 name-specific refinements, and produces 783 stable names. Refinement reduces 329 initial adjacent collisions to zero, so the observed fail-closed count is zero, while preserving every temporal mark.

The non-LLM recurrence backend counts adjacent action transitions in reference sessions and scores each transition by normalized pointwise mutual information (NPMI),

$$\text{NPMI}(a, b) = \frac{\ln[p(a, b)/(p_L(a)p_R(b))]}{-\ln p(a, b)},$$

where all probabilities use the same transition sample space (Bouma 2009). Occurrence-weighted one-dimensional k -means with $k = 2$ initializes at the minimum and maximum scores, assigns distance ties to the lower center, and uses the converged midpoint as the cutoff (MacQueen 1967). The same deterministic procedure is applied to action-changing transitions. Same-action pairs use the global cutoff; different-action pairs use the smaller cutoff. A detailed (`action`, `action_detail`) arm may remove a coarse boundary but never add one; missing or unseen detail falls back to the coarse decision.

RQ1 Details

The Git case’s 735 source nodes comprise 3 sessions, 3 prompts, 240 LLM calls, and 489 profiled tool operations.

System-effect alignment. For RQ1’s controlled cross-layer test, AgentSight first matches each timestamped process, file, or network event to the same-PID process instance whose lifetime contains it. It then links that process to a recorded tool call by an explicit tool/event ID when available, or by recorded root-process identity and overlapping process/tool lifetimes; descendants are accepted only within that tool window. The matched wrapper tool identifies the task-level semantic interval, while unmatched events remain unassigned. Across 20 real Codex tasks, this rule attributes 1,520 of 1,574 scoped process/file effects (100% precision, 96.57% recall) and rejects all 1,629 concurrent-control effects. AGENTPROF then folds exactly those joined events as unit-weight operations, preserving all 1,520 samples and every task-category total.

On AgentRewardBench, ranking operations by count versus tokens yields mean Kendall’s tau-b 0.886 with 95% interval [0.857, 0.915] over the 77 tasks with at least three distinct operations. Pooled ranking agrees at tau-b 0.929. The 13% of tasks (10 of 77) where agreement falls below 0.7 are exactly where multi-measure profiling adds value.

RQ2 Protocol

AgentRewardBench pairing enumerates every failed–successful combination within each task. The 202 distinct successful and 238 distinct failed trajectories are therefore reused when a task has multiple runs, yielding 338 pair occurrences over the 125 mixed-outcome tasks. The signed aggregate is pair-occurrence weighted and contains all 338 failed sides and all 338 successful sides.

Within each workload, operations, target-blind paths, benchmark judge/localizer predictions, and scoring rules are fixed before test targets are loaded. AgentProcessBench averages frozen judge votes within a group. HINTBench and TraceElephant assign each root-to-frame prefix the 95% Wilson lower bound of its frozen positive-prediction fraction; an operation receives the maximum score over prefixes containing it. The released HINTBench snapshot contains 536 of the paper-reported 629 trajectories. A separate 80-trajectory validation snapshot selects one of 24 field orders before the 536 test trajectories are scored.

The three localization workloads contain 1,756 trajectories. Their 1,234 queries included in MAP contribute 614,

400, and 220 AP values, respectively, while the other 522 trajectories without an annotated faulty operation remain in the population inputs. AP equals reciprocal rank for a query with one relevant operation and also handles queries with multiple relevant operations. MAP is the arithmetic mean over these queries because AP is undefined without a relevant item. The reported Direct+AGENTPROF-minus-Direct-only intervals use 10,000 paired resamples of trajectory clusters within benchmark-defined strata. AgentProcessBench uses seed 20260723, HINTBench uses 20260823, and TraceElephant uses 20260923.

The 95% confidence intervals for Direct+AGENTPROF-minus-Direct-only are [0.024, 0.039], [0.093, 0.120], and [0.088, 0.148] for the three workloads respectively. For the reading study, paired raw-minus-semantic differences are +.010 [−.021,+.042] MAP and +.120 [+.103,+.137] content fraction.

For the TraceElephant reading study, the Grok 4.5 CLI reader is invoked with tools and subagents disabled. Full-trace reading uses one single-turn call per query. Profile-guided reading first exposes only the target-blind operation skeleton; the reader selects at most five groups, after which a second single-turn call receives only those groups’ source-visible evidence. Unranked operations are appended in original order. Semantic and raw-action conditions differ only in the operation names used by the skeleton. The observed reduction from 65% to 53% is a paired difference of 0.120, or 12 percentage points.

RQ3 Protocol

For CodeTraceBench, all 405 annotations are materialized before the 2,948 human stage labels are loaded by the scorer. Codex receives prompts, commands, and output summaries, but no label or score. It emits 4,496 sparse marks over 17,148 source-native steps. CodeTraceBench stage labels define trajectory-local temporal groups, so B^3 scores the resulting operation partition and exact adjacent-boundary precision, recall, and F1 score the transition decisions. Paired task-cluster intervals preserve the benchmark’s task grouping. Literal-name accuracy is evaluated separately on the complete AgentBoard and TraceView/ASE populations.

The 95% paired task-cluster intervals for Codex-minus-baseline are [0.087, 0.116] for B^3 F1 and [0.193, 0.235] for boundary F1. The stateful Qwen2.5-3B baseline that makes push/replace/stay/pop decisions reaches 0.650 B^3 F1 and 0.257 boundary F1. Boundary error breakdown: recall 0.626, precision 0.389.

The OSWorld-Human study uses all 287 sessions, containing 3,978 operations, 3,691 adjacent pairs, and 2,042 human groups. The supervised Bernoulli Naive Bayes predictor (McCallum and Nigam 1998) fixes its model family, nine input fields, adjacent-pair features, and training procedure. Five session-held-out folds fit parameters and the decision threshold on training sessions and predict each held-out session exactly once. Label-free recurrence uses only visible transitions from the other four folds. Reference-calibrated recurrence additionally fits one scalar NPMI cutoff using the training-fold group annotations; it never reads held-out groups.

Detailed OSWorld-Human results: supervised Naive Bayes reaches 0.739 boundary F1 / 0.816 B^3 F1; calibrated recurrence reaches 0.734 / 0.801; no-LLM recurrence segmenter reaches 0.680 / 0.786 (strongest control: 0.645 / 0.678). On Mind2Web (9 sessions) and ScienceWorld (100 sessions), unsupervised TF-IDF/K-Means reaches V-measure 0.557 and 0.815 respectively. The same Codex instruction without adaptation uses coarser task-level boundaries (0.448 B^3 F1 on OSWorld-Human).

The evaluation-only Qwen3.6-27B llama.cpp backend classifies all 1,012 AgentBoard goals from goal text and the nine declared family descriptions. Three complete runs produce identical assignments. The action-label adapter uses the fixed eight TraceView action definitions and the current thought/action only; two complete runs over all 2,737 labels produce identical assignments. Literal labels use accuracy and macro-F1, partitions use V-measure or ordinary B^3 , and transitions use exact adjacent-boundary precision, recall, and F1.

AgentBoard goal classification: 0.695 macro-F1 and 0.733 accuracy (majority baseline 0.044 / 0.248). Action-label classification on AutoCodeRover, OpenHands, and RepairAgent trajectories: 0.498 macro-F1 and 0.628 accuracy (majority baseline 0.061 / 0.323). The 0.437 macro-F1 gain has a 95% bootstrap interval of [0.380, 0.494]. The fixed runtime is llama.cpp version 9870 at revision 2d973636e with the Qwen3.6-27B Q4_K_M artifact. It uses one 4,096-token slot, full GPU offload, Jinja templates, reasoning disabled, and both RAM cache and prompt reuse disabled.

Runs, Uncertainty, and Computing Environment

The CodeTraceBench direct backend completed 405 trajectories in 415 calls. The ten calls after each trajectory’s first include format retries and one authorized repair call for trajectory 53. All 405 final annotations pass the same schema validation. RQ2 fixes one profile and one benchmark prediction per trajectory, then computes uncertainty with 10,000 paired resamples. The TraceElephant reader makes one call per stage and query. OSWorld-Human uses five held-out folds. The closed-label repeat counts are given above. RQ4 construction times are medians of three complete runs; peak RSS is the largest observed semantic-profile peak. Once marks are fixed, projection, folding, serialization, and pprof readback are deterministic.

The RQ4 construction measurements use agentpprof 0.2.37 on a 24-core Intel Core Ultra 9 285K with 125 GiB RAM and Linux 6.15.11. The 27,765-operation union completes semantic construction in 1.16 s at 465.2 MiB peak RSS, versus 0.97 s for raw-action grouping. This observed range spans 729 to 27,765 operations, with 23,935 operations/s at the union scale.

The automatic CodeTraceBench pass consumes 12,050,384 input tokens, including 6,008,320 cached tokens, and 231,886 output tokens. These totals average 29,754 input and 573 output tokens over 405 trajectories. Up to four isolated workers complete 8,689.405 s of summed backend requests in 2,215.858 s of active wall time, and the deterministic downstream pipeline takes 11.516 s.

The separate AgentRewardBench pass processes 440 trajectories in 12 outcome-blind batches on a fixed two-worker schedule. It completes in 3,521.621 s of end-to-end wall time and consumes 12,039,417 input tokens, including 10,929,408 cached tokens, and 312,433 output tokens. These totals average 27,362 input and 710 output tokens per trajectory. All 12 batches pass profile validation. Deterministic materialization then takes 0.26 s for operation weights and 0.25 s for token weights. The token counters separate cached input, uncached input, output, and reasoning output independently of provider pricing.

Privacy, Ethics, and Responsible Use

Agent histories can contain prompts, model responses, source code, file paths, commands, network targets, and system-effect records. The evaluation keeps local case-study histories local and reports aggregate measurements and task-level paths. The reproduction package uses the cited public populations.

Automatic annotation applies fixed preview bounds and supports field minimization, credential and personal-data filtering, and approved provider-retention policies. Local model serving and the non-LLM recurrence backend provide local-processing options. Profiles, annotations, and intermediate packets receive the same access control and retention policy as their source traces.

The intended use is authorized engineering inspection, resource analysis, and source-linked debugging. Operators scope capture to systems and trajectories they are authorized to observe, and human review remains part of operational access-control and safety decisions.

The reported benchmark results characterize the named populations. Reusing fixed annotations avoids repeated inference, while the non-LLM backend offers a deterministic alternative.

Reproducibility Checklist Notes

Checklist items concerning novel datasets are not applicable because the paper introduces no benchmark dataset. The paper’s theoretical contribution is the semantic operation stack model, specified through the Design section’s formal definitions. Its claims require neither theorems nor proofs. This appendix supplies the final algorithm settings, target-blind freezing rules, evaluation metrics, run counts, and measured RQ4 environment. The accompanying code-and-data package supplies preprocessing and experiment source.